

Assignment 4 : Abalone Dataset – Complete Practical Guide

Aim

To predict the number of rings and age of abalone using Machine Learning algorithms such as Random Forest Classification and Linear Regression.

Objectives

1. Load and preprocess the Abalone dataset.
2. Convert categorical data into numerical form.
3. Visualize important dataset features.
4. Predict the number of rings using Classification.
5. Predict age using Linear Regression.
6. Evaluate model performance using ML metrics.

Theory

Abalone is a type of sea snail. The age of abalone is determined by counting the number of rings present in its shell. Machine Learning models can predict rings and age using physical measurements such as length, diameter, height, and shell weight. In this practical: Random Forest Classifier is used for ring prediction. Linear Regression is used for age prediction. Data visualization helps understand feature relationships.

Dataset Features

Sex Length Diameter Height Whole Weight Shucked Weight Viscera Weight Shell Weight Rings

Important Formula

Age = Rings + 1.5

Algorithms Used

1. Random Forest Classifier

Used for classification problems. It combines multiple decision trees for better prediction accuracy.

2. Linear Regression

Used for predicting continuous numerical values using a linear relationship between variables.

Evaluation Metrics

Classification:

- Accuracy Score

Regression:

- Mean Absolute Error (MAE)
- R² Score

Visualizations Included

1. Histogram
2. Heatmap
3. Scatter Plot
4. Actual vs Predicted Plot

Code Explanation

Step 1: Import required libraries.

Step 2: Load the Abalone dataset using pandas.

Step 3: Encode categorical column (Sex) using LabelEncoder.

Step 4: Visualize dataset using histograms, heatmaps, and scatter plots.

Step 5: Split dataset into training and testing sets.

Step 6: Train Random Forest model for ring prediction.

Step 7: Train Linear Regression model for age prediction.

Step 8: Evaluate performance using accuracy, MAE, and R² score.

Step 9: Predict values using user input.

How to Run .py File in Anaconda

1. Open Anaconda Prompt.
2. Install required libraries using:

```
pip install pandas numpy matplotlib seaborn scikit-learn
```

3. Navigate to project folder using cd command.

4. Run the file using:

```
python filename.py
```

How to Run .ipynb File in Jupyter Notebook

1. Open Anaconda Navigator.
2. Launch Jupyter Notebook.
3. Open the .ipynb file.

4. Run each cell one by one using Shift + Enter.

Expected Output

- Dataset information displayed successfully.
- Graphs and visualizations generated.
- Classification accuracy displayed.
- Age prediction metrics displayed.
- User input prediction performed successfully.

Outcome

Successfully implemented Machine Learning models for classification and regression using the Abalone dataset.

Conclusion

The practical demonstrates how Machine Learning techniques can predict biological characteristics using physical measurements. Random Forest provided good classification performance, while Linear Regression successfully predicted age values.

Important Viva Questions with Answers

Q1. What is the Abalone dataset?

A dataset used to predict the age of abalone using physical measurements.

Q2. Why is Label Encoding used?

To convert categorical text values into numeric form.

Q3. Why use Random Forest?

Because it provides good accuracy for classification problems.

Q4. Why use Linear Regression?

Because age prediction is a continuous numerical problem.

Q5. What is train-test split?

It divides the dataset into training and testing data.

Q6. What is overfitting?

When a model performs well on training data but poorly on unseen data.

Q7. What is R² Score?

It measures how well the regression model fits the data.

Q8. What is MAE?

Mean Absolute Error measures average prediction error.